MSc Data Science

DSM050 Data Visualisation

Midterm coursework assignment

This coursework assignment is weighted at 30% of the final mark for the module.

Introduction

Data visualisation is central to modern data science, enabling the transformation of complex datasets into meaningful graphical insights. However, the rise of AI-generated outputs (e.g., automated charts, code generation tools) introduces risks related to superficial analysis, misrepresentation, and lack of critical reasoning.

This task challenges you to move beyond AI-generated responses by demonstrating independent thinking, methodological justification, and critical evaluation in the design of visual analytics.

Objective

- Apply Python-based data preparation and visualisation techniques
- Formulate meaningful analytical questions
- Select and justify appropriate visualisation methods
- Critically evaluate visual effectiveness, bias, and limitations
- Reflect on the role and limitations of AI tools in your workflow

Midterm coursework assignment: Project

Assignment specification

You are required to select a relevant topic and identify an existing dataset that is appropriate with the chosen domain [1] You are required to formulate three clear research questions, supported by relevant literature. Using a Python-based data tool such as Matplotlib, Seaborn, or Plotly [2], perform data preprocessing, including handling missing values, manage dimensionality in your visualisation and conduct exploratory data analysis (EDA) – hence making data visualisation more efficient and effective [3]. Create visualisations to convey your findings using meaningful charts such as histograms, box plots, scatter plots, and heatmaps, to uncover trends and patterns within the dataset. Draw attention to important details using sizes, colours, fonts, and graphics. Provide context to data using visual cues and use explanatory titles to provide key insights, ensure alignment with best practices in data visualisations and perceptual design principles [4]. You are encouraged to present your visualisations as a static set of figures and include a dashboard that demonstrates clear design rationale and audience awareness.

Note: GenAI can be used for research with appropriate references, code debugging, data-visualisation prototypes with proper declaration. For example, I acknowledge the use of [1] ChatGPT (https://chat.openai.com/) to [2] generate materials for background research and self-study in the drafting of this assessment. I entered the following prompts on 4 January 2023:

> [3] Write a 50 word summary about the history of Goldsmiths, University of London. Write it in an academic style. Add references and quotations from scholarly articles.

All AI contributions must be critically evaluated and cited.

IMPORTANT: AI must not be used to generate full analytical reports, write entire sections of the final report, or fabricate data.

1. **Research Context and Problem Definition [15%]**
   - Relevant topic within a domain of interest (e.g., healthcare, finance, social sciences, environment).
   - Provide a short summary of the domain of research/field of enquiry.
   - Reference academic/technical literature where appropriate (at least 5 references to contextualise the investigation.
   - Research objective(s).
   - Target audience (e.g., policymakers, business stakeholders, general public**).**

2. **Research Questions [10%]**
   - 2-3 well-formulated research questions showing depth, focus, and feasibility.
   - Critical engagement with current and relevant literature.
   Justify why these questions matter.

3. **Dataset manipulation: Clean and preprocess the dataset using Python (e.g., Pandas) [15%]**
   - Identify an existing dataset appropriate for the chosen topic.
   - Handle missing values.
   - Prepare the data to support effective visualisation (e.g., scaling or transforming variables where necessary).

Explain how each transformation was used in your submission, not just what was done.

4. **Exploratory Data Analysis (EDA) & Visualisation [20%]**

    - Conduct EDA to summarise dataset characteristics.
    - Appropriate univariate visualisations for key variables (nominal, ordinal and numerical).
    - Key variables must be stated and visualised as univariate analysis.
    - Appropriate multivariate visualisations across a range of different data type combinations with consistent titles/figure captions and labels, attention to spacing and meaningful use of color.

5. **Interpretation and Discussion of Findings [20%]**

    - Ensure insightful interpretation of results with strong linkage to research questions and Audience.
    - Highlights implications, limitations, and patterns effectively.

6. **References, use and declaration of AI Tools [10%]**

    Evaluate the tool based on the following criteria:

    - References are accurate, complete and follow consistent style.
    - Transparent, ethical use of AI tools (e.g., ChatGPT, Colab).
    - Proper citations, reflection on limitations.
    - Clear differentiation between student and AI work.

7. **Dashboard: [10%]**

    Your audience needs at a glance understanding

    - Purposeful Design
    - Audience Alignment
    - Information Hierarchy

Submissions that rely solely on AI-generated outputs without critical engagement will receive low marks.

Submission requirements

Submit your report as a PDF file and Github repository link (approx. 2,000–2,500 words)

- The final report must be submitted in PDF format using a document editor. All visualisations, figures, and images generated as part of the analysis should be appropriately saved, labelled, and embedded within the report to illustrate key findings and support the discussion.

- You must submit a copy of the data used in your analysis, notebook (.ipynb) in Github repository.

How this Coursework will be graded

- Quality, clarity, and relevance of analysis
- Critical engagement with literature
- Correct application of methods
- Justification of choices in preprocessing and visualisation
- Quality of interpretation and discussion
- Structure, coherence, academic tone
- Correct and transparent use of AI tools

• The content within the main body of text comprises the overall word count, including in-text citations, references, quotes, heading and sub-headings. The cover page, reference list and any appendices do not count towards the overall word count.

• AI prompts can be included in the appendix. For coursework elements and the project, there is a maximum word limit. **References and appendices do not count towards the word limit.** If you exceed the word limit, we will reduce the mark you receive as follows:

Excess number of words over the word limit	Penalty applied
Up to and including 10%	5 marks deducted from original mark
More than 10% up to and including 20%	10 marks deducted from original mark
More than 20%	10 marks deducted from the original mark. The updated mark will be capped at a maximum of 40%.

References

1. https://github.com/awesomedata/awesome-public-datasets

2. Qin, X., Luo, Y., Tang, N. et al. Making data visualization more efficient and effective: a survey. The VLDB Journal 29, 93–117 (2020). https://doi.org/10.1007/s00778-019-00588-3

3. Lavanya, Addepalli, et al. "Assessing the performance of Python data visualization libraries: a review." Int. J. Computer. Eng. Res. Trends 10.1 (2023): 28-39.

4. Kelleher, Christa, and Thorsten Wagener. "Ten guidelines for effective data visualization in scientific publications." Environmental Modelling & Software
26.6 (2011): 822-827.

[END OF COURSEWORK ASSIGNMENT]